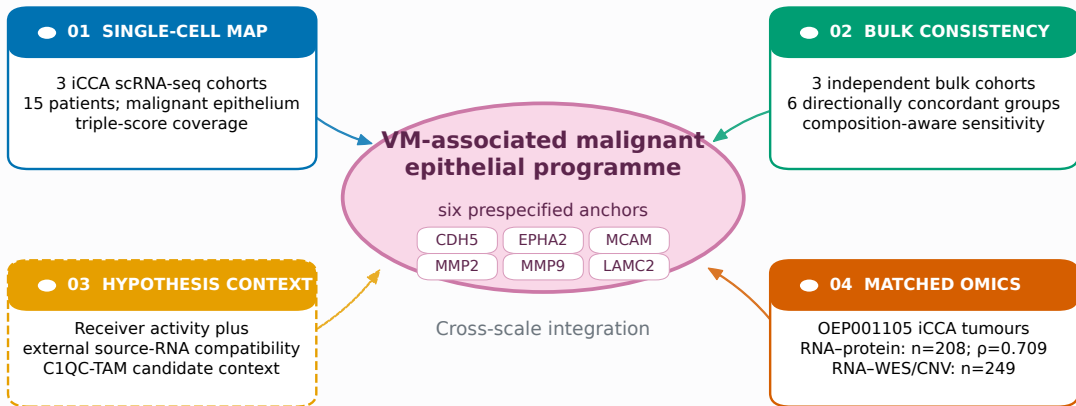


Fig. 1 | Evidence architecture of a VM-associated epithelial programme

Public iCCA cohorts are integrated in a patient-aware framework; arrows show evidence flow, not causality.



FRAMEWORK-LEVEL CONCLUSION

VM-associated epithelial programme observed across heterogeneous iCCA cohorts.
Within-project RNA-protein concordance; myeloid and genomic association-level context.

Boundary: no vasculogenic-mimicry channels/perfusion, secretion/communication, or phosphorylation activation;
no treatment resistance, independent prognosis, or drug vulnerability.